

Multi-Factor Differential Expression Analysis

Arabidopsis Ionizing Radiation Dataset: OSD-498, 502, 508, 510, 658, 782

Why this analysis is structured the way it is

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Summary — how to read the output of this script

```
knitr::opts_chunk$set(echo = TRUE, warning = TRUE, message = TRUE)
```

Why this analysis is structured the way it is

This dataset pools **six independent studies**, not six replicates of the same experiment. Before writing a single line of modeling code, three structural facts about the data dictate everything that follows:

1. **Study is a real batch effect.** Different sequencing runs, years, and likely different library preps mean “Study” itself can explain a large chunk of gene expression variance — independent of any biology. If we don’t account for this, we risk mistaking *study-to-study technical noise* for a radiation or genotype effect.
2. **Radiation type and dose are not the same variable across studies.** OSD498, OSD502, OSD508, and OSD510 all use a fixed 100 Gy dose of Co-60 gamma radiation — the only thing that varies is genotype and timepoint. OSD782 instead uses a Cs-137 source across three doses (0, 10, 100 cGy). OSD658 uses simulated Galactic Cosmic Radiation (a mixed-particle beam) at 0, 40, and 80 cGy. Lumping all of these into one “irradiated vs. not” variable would average together biologically distinct exposures.
3. **Genotype coverage is uneven.** WT is the only genotype present in all six studies. *sog1_1* appears only in OSD508 and OSD510. *myb3r135* appears only in OSD502. Any genotype comparison is necessarily scoped to the studies that actually contain that genotype.

Given these constraints, the strategy here is: **check for batch effects first, then run comparisons within the smallest sensible subset of studies that share a common design**, rather than forcing one global model across all 158 samples.

Step 1 — Install and load required packages

Reasoning: DESeq2 is the core engine for differential expression testing. limma and edgeR are included as alternates/cross-checks (useful if you later want to confirm a result isn't an artifact of one method's assumptions). pheatmap, ggplot2, and RColorBrewer are for the visual diagnostics we'll generate along the way — a PCA plot and a heatmap are how we see whether the batch-effect concern in the intro is real, rather than just asserting it.

```
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

pkgs <- c("DESeq2", "limma", "edgeR", "pheatmap", "ggplot2",
"RColorBrewer")
for (p in pkgs) {
  if (!requireNamespace(p, quietly = TRUE)) {
    if (p %in% c("DESeq2", "limma", "edgeR")) {
      BiocManager::install(p, update = FALSE, ask = FALSE)
    } else {
      install.packages(p)
    }
  }
}

library(DESeq2)
library(limma)
library(edgeR)
library(pheatmap)
library(ggplot2)
library(RColorBrewer)
```

Step 2 — Load the counts table and factors matrix

Reasoning: We deliberately split the data into two files during the cleaning phase: renamed_counts.csv (genes x samples) and factors_matrix.csv (samples x metadata). Keeping them separate — rather than encoding everything into the sample name and parsing it back out later — means every factor (Study, Genotype, Radiation, Timepoint, Replicate) is already a clean, typed column ready for a model formula. The stopifnot() check exists because **DESeq2 will silently misalign samples** if the column order in the counts matrix doesn't exactly match the row order in the factor table — this is one of the most common silent-failure modes in RNA-seq analysis, so we check it explicitly rather than trusting it.

```
counts_file <- "renamed_counts.csv"
factors_file <- "factors_matrix.csv"
```

```

counts_raw <- read.csv(counts_file, row.names = 1, check.names =
FALSE)
factors <- read.csv(factors_file, row.names = "SampleName")

# Sample names must match exactly (same set, same order) between the
two files
stopifnot(all(colnames(counts_raw) == rownames(factors)))

```

Reasoning for setting reference levels explicitly: By default, R sets a factor's reference level alphabetically. For Radiation, that would make gammaCo60_100Gy (or similar) the reference rather than none — which would flip the sign of every log2 fold-change and make results confusing to read. We explicitly force "none" and "WT" to be the reference levels so that every downstream contrast reads naturally as “effect of radiation/ mutation relative to control.”

```

factors$Study <- factor(factors$Study)
factors$Genotype <- factor(factors$Genotype, levels = c("WT",
setdiff(unique(factors$Genotype), "WT")))
factors$Radiation <- factor(factors$Radiation, levels = c("none",
setdiff(unique(factors$Radiation), "none")))
factors$Timepoint <- factor(factors$Timepoint)
factors$Replicate <- factor(factors$Replicate)

cat("Samples:", nrow(factors), "\n")
cat("Studies:", levels(factors$Study), "\n")
cat("Genotypes:", levels(factors$Genotype), "\n")
cat("Radiation levels:", levels(factors$Radiation), "\n")

```

Step 3 — Prepare counts for DESeq2

Reasoning: DESeq2's statistical model is built around count data and expects non-negative integers — it performs its own internal normalization (median-of-ratios), so feeding it *already normalized* values is a compromise, not the ideal input. The data available here is GeneLab's **normalized** counts table, not the raw/unnormalized one. We round it and floor negative values at zero purely to satisfy DESeq2's input requirements.

This is a known limitation of the current input, not a hidden step. If the raw/unnormalized counts tables become available later (*_RSEM_Unnormalized_Counts.csv or *_STAR_Unnormalized_Counts.csv, referenced inside each study's assay file), re-running this analysis on those is the more statistically correct path.

```

counts_int <- round(as.matrix(counts_raw))
counts_int[counts_int < 0] <- 0

```

Step 4 — Exploratory check: does Study dominate the variance?

Reasoning: This is the step that tests the very first assumption in the intro, instead of just assuming it. We build a minimal DESeq2 object with design = ~ Study purely to variance-stabilize the data (vst()) and

visualize it. Filtering to genes with `rowSums > 10` removes genes with essentially no signal across all samples — a standard low-count filter that also speeds up the variance-stabilizing transform. The resulting PCA plot is the single most important diagnostic in this whole script: **if samples cluster primarily by Study rather than by biology, every comparison below absolutely must include Study as a covariate** — and if they don't cluster by study, that's still useful to know, since it would mean a more pooled model is defensible after all.

```
dds_explore <- DESeqDataSetFromMatrix(
  countData = counts_int,
  colData    = factors,
  design     = ~ Study
)
dds_explore <- dds_explore[rowSums(counts(dds_explore)) > 10, ]
vsd <- vst(dds_explore, blind = TRUE)

pca_data <- plotPCA(vsd, intgroup = "Study", returnData = TRUE)
p <- ggplot(pca_data, aes(PC1, PC2, color = Study)) +
  geom_point(size = 3) +
  labs(title = "PCA colored by Study (batch check)") +
  theme_minimal()
p
ggsave("pca_by_study.png", p, width = 7, height = 5)

cat("**Read this plot before trusting anything below.** If the six
studies",
     "form visibly separate clusters, that confirms Study must be a
covariate",
     "in every model – which is exactly the approach taken in Step
5.\n")
```

Step 5 — Subset-based comparisons (the recommended approach)

Reasoning for subsetting rather than one global model: Because radiation source and dose differ by study, the only biologically valid contrasts are ones where the studies being compared actually share a consistent treatment definition. Rather than write one unwieldy formula that tries to encode every interaction across all six studies (which would also be statistically underpowered for the rarer factor combinations), we split the work into four focused models — each scoped to the studies where the comparison is actually meaningful.

5a. OSD498 + OSD510 — WT radiation response over time (Co-60, 100 Gy)

Reasoning: Both studies use the identical exposure (Co-60, 100 Gy) and overlapping timepoints, with OSD510 also contributing a `sog1_1` arm. The model formula `~ Study + Genotype + Radiation` includes `Study` as a blocking term (per Step 4's diagnostic), `Genotype` so the `sog1_1` samples in OSD510 don't bias the WT-vs-radiation contrast, and `Radiation` as the term we're actually testing.

```
sub_a <- factors[factors$Study %in% c("OSD498", "OSD510"), ]
counts_a <- counts_int[, rownames(sub_a)]
```

```

dds_a <- DESeqDataSetFromMatrix(
  countData = counts_a,
  colData    = sub_a,
  design     = ~ Study + Genotype + Radiation
)
dds_a <- dds_a[rowSums(counts(dds_a)) > 10, ]
dds_a <- DESeq(dds_a)

res_a <- results(dds_a, contrast = c("Radiation", "gammaCo60_100Gy",
"none"))
res_a <- res_a[order(res_a$padj), ]
write.csv(as.data.frame(res_a),
"DEG_OSD498_510_radiation_effect.csv")
cat("Genes at padj < 0.05:", sum(res_a$padj < 0.05, na.rm = TRUE),
"\n")

```

5b. OSD508 + OSD510 — does genotype change the radiation response?

Reasoning: This is the first model where we test an **interaction term**, not just a main effect. sog1_1 is a DNA-damage-response mutant — the scientifically interesting question isn't "is sog1_1 different from WT" or "does radiation change expression" in isolation, but whether **the size of the radiation effect itself depends on genotype**. That's exactly what Genotype:Radiation tests. We pull out the specific interaction coefficient (Genotypesog1_1.RadiationgammaCo60_100Gy) rather than a main-effect term, since that coefficient name is what DESeq2 generates for the interaction in this particular factor encoding.

```

sub_b <- factors[factors$Study %in% c("OSD508", "OSD510"), ]
counts_b <- counts_int[, rownames(sub_b)]

dds_b <- DESeqDataSetFromMatrix(
  countData = counts_b,
  colData    = sub_b,
  design     = ~ Study + Genotype + Radiation + Genotype:Radiation
)
dds_b <- dds_b[rowSums(counts(dds_b)) > 10, ]
dds_b <- DESeq(dds_b)

# Does the radiation response differ between WT and sog1_1?
res_b_interaction <- results(dds_b, name =
"Genotypesog1_1.RadiationgammaCo60_100Gy")
res_b_interaction <-
res_b_interaction[order(res_b_interaction$padj), ]
write.csv(as.data.frame(res_b_interaction),
"DEG_OSD508_510_genotype_x_radiation_interaction.csv")
cat("Genes showing genotype-dependent radiation response (padj <
0.05):",
    sum(res_b_interaction$padj < 0.05, na.rm = TRUE), "\n")

```

5c. OSD782 — Cs-137 dose-response, WT only

Reasoning: This study is internally self-contained (one radiation source, one genotype), so there's no need for a Study covariate here — the only structure to account for is Timepoint, since the same dose can have a very different effect at 1h vs. 72h post-exposure. We run **two separate contrasts** (10 cGy vs. none, and 100 cGy vs. none) rather

than treating dose as a continuous/ordinal term, since with only three dose levels and a likely non-linear dose-response curve, discrete contrasts are more interpretable than forcing a linear dose coefficient.

```
sub_c <- factors[factors$Study == "OSD782", ]
counts_c <- counts_int[, rownames(sub_c)]

dds_c <- DESeqDataSetFromMatrix(
  countData = counts_c,
  colData = sub_c,
  design = ~ Timepoint + Radiation
)
dds_c <- dds_c[rowSums(counts(dds_c)) > 10, ]
dds_c <- DESeq(dds_c)

res_c_low <- results(dds_c, contrast = c("Radiation",
"gammaCs137_10cGy", "none"))
res_c_high <- results(dds_c, contrast = c("Radiation",
"gammaCs137_100cGy", "none"))
write.csv(as.data.frame(res_c_low[order(res_c_low$padj), ]),
"DEG_OSD782_10cGy_vs_none.csv")
write.csv(as.data.frame(res_c_high[order(res_c_high$padj), ]),
"DEG_OSD782_100cGy_vs_none.csv")
cat("10 cGy vs none, genes at padj < 0.05:", sum(res_c_low$padj <
0.05, na.rm = TRUE), "\n")
cat("100 cGy vs none, genes at padj < 0.05:", sum(res_c_high$padj <
0.05, na.rm = TRUE), "\n")
```

5d. OSD658 — simulated Galactic Cosmic Radiation dose-response, WT only

Reasoning: Same logic as 5c, but there's no Timepoint factor at all in this study (single harvest time), so the design formula simplifies to just ~ Radiation. This is also the only study using a mixed-particle beam rather than a pure gamma source — worth keeping mentally separate from 5a/5b/5c when interpreting results, since “radiation response” here may reflect a biologically different stress (particle radiation causes more clustered DNA damage than gamma rays).

```
sub_d <- factors[factors$Study == "OSD658", ]
counts_d <- counts_int[, rownames(sub_d)]

dds_d <- DESeqDataSetFromMatrix(
  countData = counts_d,
  colData = sub_d,
  design = ~ Radiation
)
dds_d <- dds_d[rowSums(counts(dds_d)) > 10, ]
dds_d <- DESeq(dds_d)

res_d_40 <- results(dds_d, contrast = c("Radiation", "GCR_40cGy",
"none"))
res_d_80 <- results(dds_d, contrast = c("Radiation", "GCR_80cGy",
"none"))
write.csv(as.data.frame(res_d_40[order(res_d_40$padj), ]),
"DEG_OSD658_GCR40_vs_none.csv")
write.csv(as.data.frame(res_d_80[order(res_d_80$padj), ]),
"DEG_OSD658_GCR80_vs_none.csv")
cat("GCR 40cGy vs none, genes at padj < 0.05:", sum(res_d_40$padj <
0.05, na.rm = TRUE), "\n")
```

```
cat("GCR 80cGy vs none, genes at padj < 0.05:", sum(res_d_80$padj <
0.05, na.rm = TRUE), "\n")
```

Step 6 — Full pooled model (exploratory visualization only)

Reasoning — and an explicit caveat: This step deliberately builds a single model across all 158 samples, but **only to generate an overview heatmap**, not to report a formal p-value. The Radiation factor isn't comparable across all six studies (gamma vs. Cs-137 vs. mixed-particle GCR, at different doses), so testing "radiation main effect" in a pooled model would average together biologically distinct exposures and produce a result that's hard to defend in any write-up. The model here only includes Study + Genotype — explicitly *not* Radiation — for that reason. What this step is good for: seeing, at a glance, which genes vary the most across the whole dataset and whether that variation tracks with Study, Genotype, or something else entirely (worth a visual sanity check even if it won't appear in the final results).

```
dds_full <- DESeqDataSetFromMatrix(
  countData = counts_int,
  colData    = factors,
  design     = ~ Study + Genotype
)
dds_full <- dds_full[rowSums(counts(dds_full)) > 10, ]
vsd_full <- vst(dds_full, blind = FALSE)

topvar <- head(order(rowVars(assay(vsd_full))), decreasing = TRUE),
50)
annotation_col <- factors[, c("Study", "Genotype", "Radiation")]

pheatmap(assay(vsd_full)[topvar, ],
  annotation_col = annotation_col,
  show_rownames = FALSE,
  show_colnames = TRUE,
  fontsize_col  = 5,
  main = "Top 50 variable genes across all 6 studies")
```

Summary — how to read the output of this script

1. **Look at the PCA plot from Step 4 first.** If clusters separate cleanly by Study, every subset model above is justified in treating Study as a covariate (it already does).
2. **Trust the subset comparisons (5a-5d) over the pooled model (Step 6)** for any actual reported result — the pooled model is for visualization only, as explained above.
3. **5b's interaction term is the most scientifically interesting result** in this script: a significant Genotype:Radiation interaction means *sog1_1* doesn't just have a different baseline expression profile, it genuinely changes *how the plant responds to radiation* — which is the kind of finding worth following up with pathway enrichment.
4. Every DEG_*.csv written by this script is a candidate input for

downstream GO/KEGG enrichment (Phase 5 of the project outline)
or for cross-checking against the same comparison run in iDEP3.0.